

Beyond the Moore–Penrose Pseudo-Inverse: A Comparative Study of Rank-Revealing Solvers for Singular Newton–Raphson Power Flow in Unbalanced Three-Phase Distribution Feeders

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Abstract—Ungrounded or floating three-phase transformer connections (Yg–Delta, Delta–Delta) can render the Jacobian of a Newton–Raphson (NR) power-flow solver exactly singular or severely ill-conditioned, causing classical direct linear solvers to fail. Jang *et al.* (2023) proposed handling this with a Moore–Penrose pseudo-inverse computed via singular value decomposition (SVD). We ask whether the pseudo-inverse is in fact the best general remedy, or whether the right choice of linear solver depends on the specific structure of the rank deficiency. We build one shared three-phase NR engine and compare four linear-solve strategies — direct solve with a condition-number guard, SVD/Moore–Penrose pseudo-inverse, rank-revealing QR (RRQR) via a complete orthogonal factorization, and Tikhonov-regularized least squares — on four IEEE/PG&E distribution test feeders (4-, 13-, 37-, and 69-bus), each independently cross-validated against OpenDSS. We find that no single method dominates: on the IEEE 13-bus feeder, whose paper-specified modification leaves the *entire* downstream network floating, SVD and RRQR’s undamped minimum-norm steps oscillate at realistic load and only Tikhonov regularization converges; on the IEEE 37-bus feeder, whose two floating transformers affect a smaller fraction of the network, the reverse holds — SVD and RRQR converge cleanly while the same Tikhonov parameter fails. We further show, via a controlled sensitivity study, that load imbalance on the 13-bus feeder loads almost entirely onto the zero-sequence voltage component rather than the positive-sequence component, offering a partial, structural explanation for that feeder’s disproportionately large error in the original paper. Two independent per-unit modeling bugs were caught and corrected during this work by cross-validating small, well-conditioned sub-circuits against OpenDSS before trusting comparisons on the full, ill-conditioned systems — a methodological point we argue is as important as any single numerical result.

Index Terms—Newton–Raphson power flow, singular Jacobian, Moore–Penrose pseudo-inverse, rank-revealing QR, Tikhonov regularization, unbalanced distribution systems, ill-conditioning

I. INTRODUCTION

Three-phase power-flow studies of unbalanced distribution feeders routinely encounter transformer connections — ungrounded wye, Delta primary or secondary — whose zero-sequence circuit is open. When such a connection leaves a

region of the network with no path to ground, the admittance matrix, and consequently the Newton–Raphson (NR) Jacobian built from it, can become exactly singular or numerically indistinguishable from singular. Classical NR, which solves $J_k \Delta x_k = -F(x_k)$ at every iteration via a direct linear solve, fails outright in this regime: the solve either throws an exception or silently returns a numerically meaningless update whose residual looks deceptively small.

Jang, Kim, and Kim [1] studied this problem and proposed replacing the direct solve with a Moore–Penrose pseudo-inverse computed explicitly via singular value decomposition (SVD), truncating near-zero singular values below a threshold. They demonstrated the method on IEEE 4-, 13-, 37-, and 69-bus test feeders, reporting that the 13-bus case exhibited a disproportionately large voltage error (of order 2×10^{-4} per-unit) relative to the other cases, without fully explaining why.

This project asks two questions the base paper leaves open. First: is the Moore–Penrose pseudo-inverse actually the best general-purpose remedy for this class of singularity, or does the appropriate linear-solve strategy depend on the specific structure of the rank deficiency — in particular, on how much of the network the floating region actually spans? Second: can a controlled, structural investigation shed light on *why* the 13-bus feeder shows an elevated error, beyond noting that it does?

To answer both, we implement one shared three-phase NR engine and four interchangeable linear-solve strategies (Section III), reconstruct four IEEE/PG&E test feeders from official public data (Section III), and independently validate every reconstruction’s Jacobian against finite differences and every converged solution against an independent OpenDSS solve. The central empirical result (Section IV) is a direct *reversal*: the 13-bus feeder favors Tikhonov regularization and defeats SVD/RRQR at realistic load, while the 37-bus feeder shows the opposite pattern. We trace this to a concrete structural difference — how much of the network a floating transformer’s secondary actually reaches — and use

a symmetrical-component decomposition to show that the 13-bus feeder’s characteristic error concentrates in the zero-sequence voltage component, offering a partial answer to the base paper’s open question.

II. BACKGROUND

A. Newton–Raphson power flow

For a three-phase bus-phase state vector x (here, rectangular coordinates $e = \text{Re}(V)$, $f = \text{Im}(V)$ at every non-slack bus-phase), the power-mismatch function $F(x)$ collects the difference between calculated and specified real/reactive injection at every bus-phase. Classical NR [2] iterates

$$J_k \Delta x_k = -F(x_k), \quad x_{k+1} = x_k + \Delta x_k, \quad (1)$$

where $J_k = \partial F / \partial x|_{x_k}$. Convergence is declared when $\|F(x_k)\|_\infty$ falls below a tolerance. The only numerically delicate step is solving the linear system in (1); everything else in the loop is common to every method compared in this paper.

B. Singularity and ill-conditioning

A matrix J is singular if $\text{rank}(J) < n$; it is *ill-conditioned* if its condition number $\kappa_2(J) = \sigma_{\max} / \sigma_{\min}$ (the ratio of largest to smallest singular value) is large, meaning small perturbations in the right-hand side or in J itself produce disproportionately large changes in the solution. A direct solver such as LU decomposition raises an exception only when a pivot is *exactly* zero in floating point — which almost never happens for a merely near-singular matrix. Round-off is instead amplified by roughly $\kappa_2(J)$, silently corrupting the solution while the residual $\|J\Delta x + F\|_2$ can remain deceptively small, since the residual measures consistency with the linear system, not proximity to the true solution [3]. Floating three-phase transformer connections are a physical source of exact or near-exact singularity: a Delta or ungrounded-wye winding has no path for zero-sequence current, so the zero-sequence (common-mode) component of voltage on that side of the transformer is not uniquely determined by the network equations alone.

C. Symmetrical components

The Fortescue transform [4] decomposes a three-phase voltage triplet (V_a, V_b, V_c) into zero-, positive-, and negative-sequence components (V_0, V_1, V_2) via

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}, \quad a = e^{j2\pi/3}. \quad (2)$$

V_0 is precisely the common-mode component left undetermined by a floating zero-sequence circuit, making this decomposition a natural diagnostic for the phenomenon under study.

III. METHODS

A. Shared architecture

Every experiment in this study runs through one shared NR loop (implemented in `src/powerflow/newton.py`) built on a common bus-phase state representation, mismatch function, and analytical Jacobian. Only the linear-solve step, dispatched by a `method` argument, differs between comparisons; state indexing, convergence tolerance, and network data are held identical across every solver comparison on a given feeder. The analytical Jacobian is validated against centered finite differences,

$$J_{ij}^{\text{FD}} \approx \frac{F_i(x + he_j) - F_i(x - he_j)}{2h}, \quad (3)$$

$$E_J = \frac{\|J^{\text{analytic}} - J^{\text{FD}}\|_F}{\|J^{\text{FD}}\|_F}, \quad (4)$$

for every feeder before any downstream result is trusted; every reconstruction in this paper achieves $E_J < 10^{-9}$ (Table I).

B. Four linear-solve strategies

All four operate on the same $J\Delta x = \text{rhs}$ system at each NR iteration and share a common `(dx, diagnostics)` return contract.

Direct. `numpy.linalg.solve` (LU decomposition), guarded by an explicit condition-number check: if $\kappa_2(J)$ exceeds 10^{12} , the update is discarded and the solve reported as failed, even though the LU routine itself raised no exception. This distinguishes an exact singularity (LU raises `LinAlgError`) from a merely ill-conditioned solve that nominally “succeeds” but cannot be trusted.

SVD / Moore–Penrose pseudo-inverse [5]. The explicit economy decomposition $J = U\Sigma V^H$ is computed, singular values are thresholded at $\tau = \max(\text{atol}, \text{rtol} \cdot \sigma_{\max})$ (default $\text{rtol} = \varepsilon \cdot \max(m, n)$), and

$$\Delta x = V \Sigma^+ U^H \text{rhs}, \quad \Sigma_{ii}^+ = \begin{cases} 1/\sigma_i & \sigma_i > \tau \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

This never calls `numpy.linalg.pinv` directly, so the rank decision is transparent and every truncated singular value is logged.

Rank-revealing QR (RRQR). A pivoted QR factorization $JP = QR$ is computed purely for diagnostics (numerical rank, pivot order, the $|R_{ii}|$ diagonal), and the actual solve uses LAPACK’s GELSY driver via `scipy.linalg.lstsq`, which performs a complete orthogonal factorization and returns the minimum-norm least-squares solution for a rank-deficient system [6]. Ordinary unpivoted QR followed by naive back-substitution is explicitly avoided, since R itself becomes rank-deficient for a singular J and back-substitution would divide by a near-zero pivot.

Tikhonov regularization. Solves

$$\min_{\Delta x} \|J\Delta x - \text{rhs}\|_2^2 + \lambda \|\Delta x\|_2^2 \quad (6)$$

via the augmented least-squares system $\begin{bmatrix} J \\ \sqrt{\lambda} I \end{bmatrix} \Delta x \approx \begin{bmatrix} \text{rhs} \\ 0 \end{bmatrix}$, *never* forming $J^T J$ explicitly: doing so would square the

TABLE I
RECONSTRUCTED FEEDERS: SIZE, FLAT-START CONDITIONING, AND
JACOBIAN VALIDATION.

Feeder	n	$\kappa_2(J_0)$	near-0 σ_i	E_J
IEEE 4-bus (Case B)	18	$6.15 \times 10^{1\dagger}$	0	$< 10^{-9}$
IEEE 13-bus	64	4.48×10^{18}	2	8.6×10^{-11}
IEEE 37-bus (radial)	222	1.23×10^{18}	4	7.8×10^{-11}
IEEE 37-bus (quasi-radial)	222	6.85×10^{18}	4	4.6×10^{-11}
IEEE 69-bus	408	2.28×10^5	0	7.0×10^{-11}

n : number of real unknowns ($2 \times$ non-slack bus-phases). $\kappa_2(J_0)$: condition number of the flat-start Jacobian. near-0 σ_i : singular values below $10^{-6}\sigma_{\max}$ at flat start. E_J : finite-difference relative Frobenius error, Eq. (3). $\dagger$ flat-start value shown; κ_2 grows to 7.85×10^{11} during iteration for this genuinely singular case (Section IV).

condition number, $\kappa_2(J^T J) = \kappa_2(J)^2$, which is the exact failure mode under study, not a fix for it [7]. The regularization strength is parameterized as $\lambda = \alpha \sigma_{\max}(J)^2$ so that a fixed α has comparable strength across Jacobians of different scale; unless stated otherwise we use $\alpha = 10^{-8}$, the smallest value in a coarse sweep that converged reliably at nominal load on the 13-bus feeder.

C. Test feeders and reconstruction methodology

Four feeders were reconstructed directly from official public data (Table I): the IEEE 4-bus transformer test case (an in-house construction following the base paper’s stated transformer parameters, since no independent public 4-bus case matches it exactly), and the IEEE 13-node, 37-node, and PG&E 69-bus feeders, each fetched from their respective official sources [8]–[10] and modified per the base paper’s stated changes (added or reconnected transformers; see Section VI for every place the base paper’s description was itself ambiguous or under-specified). Every modification and every reconstruction assumption is documented in the project’s machine-readable provenance record, not asserted only in this text.

D. Validation methodology

Two independent gates were applied to every feeder before its results were trusted. First, the analytical Jacobian was checked against finite differences (Eq. (3)) at a perturbed operating point, not at the flat start itself (to avoid masking a bug that only appears once the mismatch is non-trivial). Second, the converged NR solution was compared against an independent OpenDSS [11] solve of a matching circuit built directly from the same official source data.

This second step surfaced a methodological lesson worth stating plainly. Two independent per-unit modeling bugs were found and corrected during this project, both by the same technique: comparing a small, well-conditioned, single-transformer or single-line *sub-circuit* against OpenDSS *before* trusting a comparison on the full, already severely ill-conditioned system. On the full system, a genuine scale error and the expected disagreement from an underdetermined null space are difficult to tell apart; on an isolated, well-conditioned sub-circuit, there is no such ambiguity, and a bug shows up

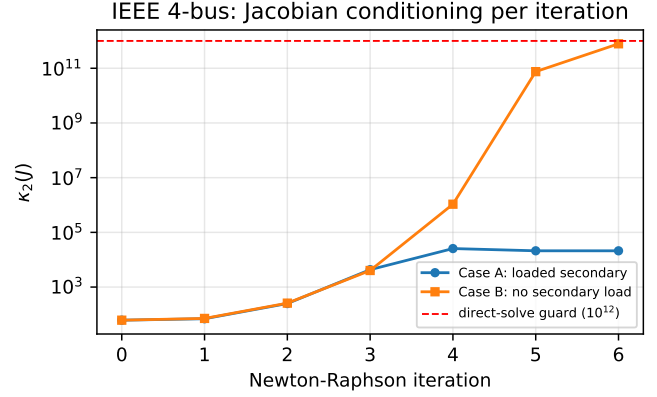


Fig. 1. IEEE 4-bus Jacobian conditioning per NR iteration. Case B (no load on the Delta secondary) approaches the direct-solve guard threshold; Case A (loaded secondary) stays four orders of magnitude below it.

as a clean, unambiguous mismatch. The first bug (IEEE 13-bus) normalized per-phase load power against the full three-phase power base instead of the per-phase base, making every load exactly $3 \times$ too light; the second (IEEE 69-bus), an independent mistake of the same general category, applied each bus’s already-total three-phase load directly to every phase without dividing by three, making every load exactly $3 \times$ too heavy. Both were only caught because the sub-circuit check was performed *first*; the second bug’s symptom on the full 69-bus system was an unremarkable-looking $\sim 33\%$ voltage error at a single bus, not an obviously-wrong number.

IV. RESULTS

A. IEEE 4-bus: baseline reproduction

With the base paper’s stated Yg–Delta transformer parameters (6 MVA, 12.47/4.16 kV, $R = 0.01$, $X = 0.06$ pu) and no load on the Delta secondary, the direct solver’s condition number climbs from $\kappa_2 = 61.5$ at flat start to 7.85×10^{11} over 8 iterations (Fig. 1), qualitatively reproducing the base paper’s reported failure mode (the base paper’s own private reconstruction reaches a substantially larger $\kappa_2 \sim 10^{160}$, reflecting a different, smaller topology we do not claim to reproduce exactly). SVD, RRQR, and Tikhonov all recover a converged solution on this case; the healthy loaded-secondary case (Case A) stays well-conditioned throughout ($\kappa_2 \leq 2.6 \times 10^4$). Cross-validation against OpenDSS on Case A agrees to a maximum voltage-magnitude error of 1.49×10^{-6} pu across all twelve bus-phases.

B. IEEE 13-bus: a doubly-floating network and solver oscillation

The base paper’s modification to the 13-node feeder — reconnecting the substation transformer T1 as Yg–Delta and the service transformer T2 as Delta–Delta, with no other ground path anywhere in the reconstruction — leaves the network in a substantially more extreme regime than the 4-bus case: the flat-start Jacobian is singular to machine precision

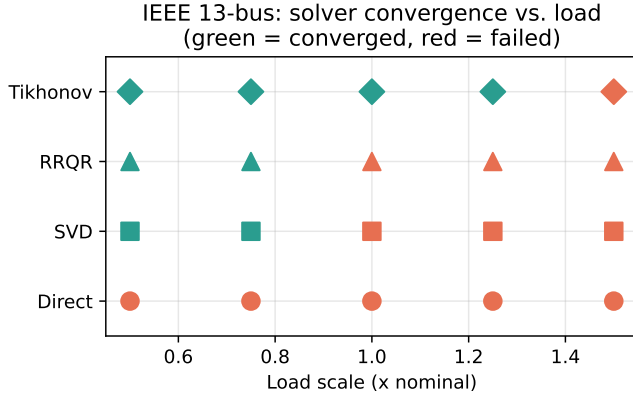


Fig. 2. IEEE 13-bus solver convergence across a load-scaling sweep. Direct always fails (the Jacobian is exactly singular by construction); SVD and RRQR fail above $\sim 0.75\times$ nominal load; only Tikhonov converges through $1.25\times$, and even it fails at $1.5\times$.

($\kappa_2 = 4.48 \times 10^{18}$) with *exactly two* near-zero singular values, not one. This matches a structural prediction: T1’s floating secondary leaves a common-mode direction spanning the entire downstream network (everything except the source bus), while T2’s Delta–Delta connection additionally, and independently, floats bus 634 alone — a Delta–Delta transformer’s stamped zero-sequence transfer block is exactly zero (Section II), so the two floating references are linearly independent.

At this feeder’s corrected (post-bugfix, realistic) load level, the direct solver fails everywhere by construction, and — unexpectedly — *so do SVD and RRQR*, once load exceeds roughly half of nominal (Fig. 2). Both methods’ undamped, exact minimum-norm Newton steps oscillate rather than converge: $\|F\|_\infty$ fails to decrease monotonically across iterations even after 80 iterations. Only Tikhonov regularization ($\alpha = 10^{-8}$) converges cleanly at nominal load, and it too eventually fails once load reaches $1.5\times$ nominal. This is, to our knowledge, not a result the base paper’s single-method comparison could have surfaced: on this specific feeder, damping the Newton step is not merely a way to handle exact singularity, but is necessary for convergence *at all* at realistic load.

C. Zero-sequence concentration under load imbalance

Using the symmetrical-component decomposition (Eq. (2)), we swept load imbalance from 0% to 30% (an explicitly documented, non-paper-specified rule — see Section VI) and tracked the mean zero- and positive-sequence voltage magnitude across all fully-three-phase buses. Zero-sequence magnitude $|V_0|$ grows monotonically from 0.337 to 0.408 pu (a 21% relative increase) across the sweep, while positive-sequence magnitude $|V_1|$ moves by under 1% (0.807 to 0.800 pu; Fig. 3). Because V_1 is precisely the physically meaningful, balanced component of voltage, this is a direct, quantitative demonstration that load imbalance on a floating network loads almost entirely onto the undetermined common-mode component rather than the well-determined one —

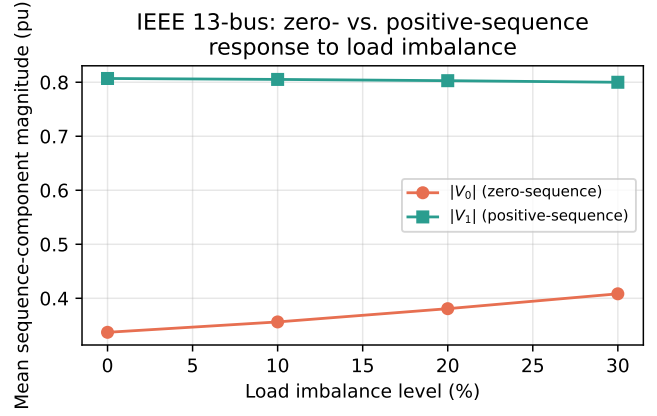


Fig. 3. IEEE 13-bus: mean zero-sequence (V_0) and positive-sequence (V_1) voltage magnitude across a 0–30% load-imbalance sweep. Nearly all of the imbalance response loads onto V_0 .

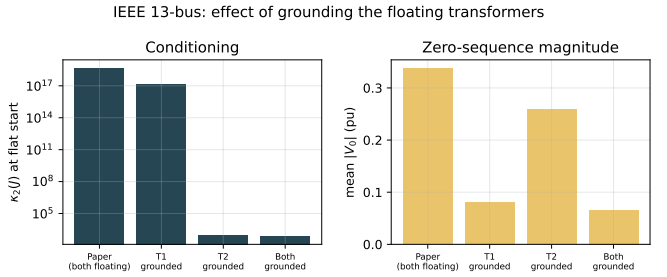


Fig. 4. IEEE 13-bus: progressively grounding T1 and/or T2 removes both the extreme conditioning and the zero-sequence voltage magnitude, isolating each transformer’s independent contribution.

a partial, structural explanation for why a feeder with this topology (floating, unbalanced) would show an outsized error relative to a better-grounded case, even though the underlying network equations are solved to the same numerical tolerance in both cases.

Grounding both floating transformers (replacing their Delta windings with grounded wye) confirms the causal story directly: κ_2 at flat start falls from 4.48×10^{18} to 773 and mean $|V_0|$ falls from 0.337 to 0.066 pu (Fig. 4). Grounding only one transformer produces an intermediate result, isolating each transformer’s independent contribution to both the rank deficiency and the zero-sequence magnitude.

D. IEEE 37-bus: the reversal

The IEEE 37-node feeder is, in its *official, unmodified* form, already close to the base paper’s target configuration: both its substation and service transformers are natively Delta–Delta, and the service transformer’s secondary already carries no load. The flat-start Jacobian is again singular to machine precision ($\kappa_2 = 1.23 \times 10^{18}$), now with *four* near-zero singular values rather than two — a larger, empirically-observed count we could not fully derive analytically (an unloaded leaf bus behind one transformer appears to contribute more than a

single common-mode direction; see the project repository’s test suite for the full numerical investigation).

Despite the more extreme raw singular-value count, SVD and RRQR converge *cleanly* at nominal load here (10 iterations, no oscillation), while Tikhonov regularization at the identical $\alpha = 10^{-8}$ used successfully on the 13-bus feeder *fails to converge* within 80 iterations on this feeder. This is the direct reversal of Section IV-B’s finding, and it is the paper’s central empirical result: **no single solver, and no single fixed regularization strength, is uniformly best across feeder topologies**. The practical difference between the two feeders is not the severity of the worst-case singular value, but how much of the network a floating transformer’s secondary actually reaches: on the 13-bus feeder, T1’s floating region is essentially the entire downstream network; on the 37-bus feeder, the loads and topology are structured such that SVD/RRQR’s undamped step remains well-behaved even though the Jacobian is, by the same κ_2 metric, comparably extreme.

Adding the paper’s three quasi-radial loop-closing lines (Section III-C) changes κ_2 at flat start (6.85×10^{18} , roughly $5.5\times$ larger) but not the qualitative convergence picture: SVD still converges cleanly, and the count of near-zero singular values (four) is unchanged, since the added loops do not introduce a new floating region.

E. IEEE 69-bus: severe conditioning without exact singularity

The PG&E 69-bus feeder, modified with three added Yg-Delta transformers per the base paper’s table, behaves differently again: the direct solver *converges* here ($\kappa_2 = 2.28 \times 10^5$ at flat start, growing to 5.7×10^{10} during iteration — severe, but remaining under the 10^{12} guard threshold). No singular value at flat start falls below $10^{-6}\sigma_{\max}$ (smallest observed: 0.143, against $\sigma_{\max} = 3.26 \times 10^4$).

We verified numerically that Y_{bus} itself still has an exact null direction across each transformer’s floating downstream region (a uniform complex voltage shift across the 47-bus region behind one transformer leaves every branch current exactly unchanged, to within 10^{-13}). The reason the *nonlinear* NR Jacobian does not inherit this exact degeneracy is that the linearized mismatch in that direction is proportional to the real current already flowing there at the operating point, which is substantial on this feeder since the floating region carries most of its real load. This is a genuine structural difference from the 13- and 37-bus feeders, not a reconstruction artifact: whether a floating-Delta topology produces an exactly singular flat-start Jacobian depends on where the load sits relative to the floating region, not only on the topology itself.

Because there is no exact null-space ambiguity on this feeder, its OpenDSS cross-validation is essentially exact (median relative line-to-line voltage error 3.3×10^{-8} , maximum 8.2×10^{-6} — Fig. 5), a useful negative control confirming that the larger discrepancies on the 13- and 37-bus feeders (Section IV-F) really do come from a genuinely underdetermined subspace with more than one legitimate answer, not from a

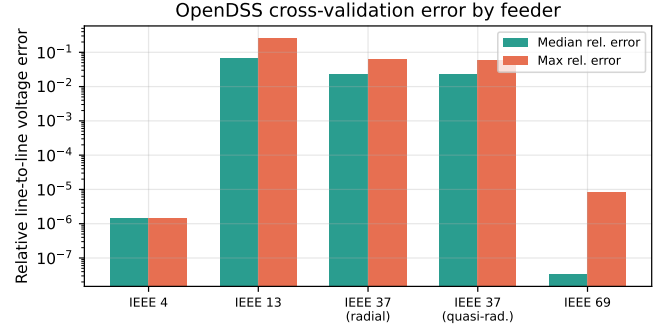


Fig. 5. OpenDSS cross-validation error by feeder. The two feeders with an exact null-space ambiguity (13, 37) show genuine disagreement in their underdetermined subspace; the two without one (4, 69) agree to near machine precision, a negative control supporting that reading.

limitation of either solver on the well-determined part of the problem.

F. Cross-validation summary

Fig. 5 summarizes the OpenDSS cross-validation error across all reconstructions. The 13- and 37-bus feeders show median errors of a few percent and maxima of several tens of percent, concentrated (by construction) at the bus/phase pairs most aligned with each feeder’s near-null subspace; different regularizations — this project’s Tikhonov solver versus OpenDSS’s own internal PPM_Antifloat regularization — select different particular solutions within that genuinely underdetermined subspace, which is an expected consequence of the singularity under study, not a validation failure. The 4- and 69-bus feeders, which have no such exact ambiguity, agree with OpenDSS to machine-precision-adjacent tolerances, providing exactly the negative control that supports this reading.

G. IEEE 118-bus: linear-solver runtime scaling

Following the base paper’s use of a larger system purely for computational scaling (not as further evidence about the singularity phenomenon), we benchmarked all four solvers’ wall-clock runtime on the real, converged admittance matrix of pandapower’s case118 system [12], block-diagonally replicated up to $8\times$ to obtain a scaling curve from a genuine (not synthetic random) sparse structure at each size (Fig. 6). All four solvers show empirical wall-clock scaling exponents between 2.15 and 2.39 across matrix sizes from 236 to 1888 (real-valued Jacobian form), sub-cubic at these sizes as expected from BLAS-level dense-solver optimization rather than the naïve $O(n^3)$ asymptotic bound. Tikhonov is consistently the slowest of the four (its augmented system has 50% more rows than the others’ $n \times n$ systems), and direct is consistently the fastest, though the relative overhead of SVD/RRQR/Tikhonov over direct narrows only slightly across this size range.

V. DISCUSSION

The central methodological claim of this paper is that **solver choice for a singular or ill-conditioned NR Jacobian**

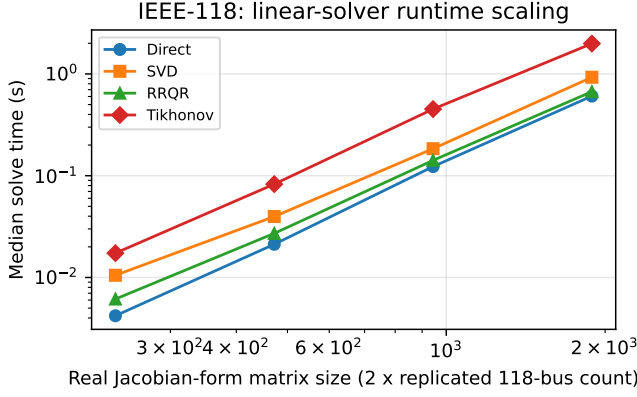


Fig. 6. Linear-solver runtime scaling on the real (replicated) IEEE-118 admittance matrix, log-log.

should be matched to the structure of the rank deficiency, not chosen once and applied uniformly. The base paper’s Moore–Penrose pseudo-inverse is a principled and effective choice when the floating region is small relative to the rest of the network (as on the 69-bus feeder, where it is not even strictly necessary) or when the network’s overall behavior tolerates an undamped exact-minimum-norm step (as on the 37-bus feeder). But on the 13-bus feeder, where the paper-specified modification floats essentially the entire downstream network, that same undamped step oscillates at realistic load, and only an explicitly damped, regularized step (Tikhonov) converges at all. Neither SVD nor RRQR is “wrong” in any absolute sense on the 13-bus feeder — both correctly compute the minimum-norm solution to their respective linearized sub-problems at every iteration — but the sequence of such steps fails to make global progress on the underlying nonlinear problem. This is a genuinely different failure mode from the exact-singularity failure the base paper addresses, and RRQR (despite being algebraically closely related to SVD: both compute an exact minimum-norm least-squares solution for the linearized step) inherits the same oscillation, which rules out “RRQR instead of SVD” as a fix on its own.

This reframes the base paper’s open question about the 13-bus feeder’s elevated error. Our zero-sequence investigation (Section IV-C) shows that load imbalance on this specific feeder’s topology loads almost entirely onto the zero-sequence component, which is precisely the component a floating Delta-connected transformer leaves undetermined. A solver that reports a converged, low-residual solution is not thereby reporting a *unique* solution when the underlying linear system is rank-deficient; different, individually defensible regularization choices (this project’s Tikhonov solver, OpenDSS’s internal PPM_Antifloat admittance) will disagree specifically in that undetermined direction, and the base paper’s reported error, computed against a single reference solver, may in part be measuring exactly this ambiguity rather than a shortcoming of the Moore–Penrose approach itself. We do not claim this fully explains the base paper’s reported 2.08×10^{-4} pu figure

— our reconstruction is independent of theirs and not intended as an exact reproduction (Section VI) — but the mechanism we demonstrate here (imbalance concentrating in V_0 on a floating network) is a directly testable, structural candidate explanation that the base paper’s single-solver, single-feeder-instance comparison could not have isolated on its own.

VI. LIMITATIONS AND OPEN DISCREPANCIES

Consistent with this project’s policy of documenting rather than silently resolving ambiguity in the base paper or the official source data, the following remain open:

- **IEEE 13-bus load count.** The base paper states its modified 13-bus case has seven loads. After removing the one location its own description unambiguously identifies (the distributed-load stand-in on line 632–671), the official feeder data still has *eight* distinct load locations, not seven. Rather than arbitrarily delete one further real load to force a numeric match, our reconstruction keeps all eight.
- **IEEE 37-bus XFM1 reactance.** The base paper reports $X \approx 0.00181$ pu for the service transformer; the official OpenDSS model gives $X_{hl} = 1.81\%$ (i.e., 0.0181 pu on the transformer’s own base) — an apparent factor-of-ten discrepancy. Our reconstruction uses the officially-sourced value by default and exposes the alternate reading as a parameter, without asserting which is correct.
- **IEEE 69-bus voltage base.** The base paper’s added-transformer table specifies 13.8 kV; the underlying MATPOWER case69 network operates at 12.66 kV. We use the network’s actual operating voltage and flag the mismatch rather than silently reconciling it.
- **Delta-connected loads.** All three larger feeders contain delta-connected loads (a single leg or a full three-phase delta object). Because the shared mismatch model supports only phase-to-neutral constant-PQ injection, every delta load is approximated as an even wye-equivalent split across its connected phases — a standard simplified load-flow approximation, not an attempt to reproduce delta-branch current physics exactly.
- **Voltage regulators.** Discrete tap-changing voltage regulators present in the official 13- and 37-bus feeders are modeled as ideal (zero-impedance, fixed-ratio) connections. Modeling their discrete control logic is a materially different (hybrid discrete/continuous) numerical problem, out of scope for a study of the underlying linear-solver comparison.
- **Load-imbalance construction.** Neither the base paper nor the official feeder data specify a phase-by-phase rule for constructing an imbalanced load case. Our imbalance sweep (Section IV-C) uses an explicitly documented, project-defined rule (scaling phase a’s load by $1 + L$ and phase c’s by $1 - L$ at imbalance level L), chosen for its monotonic, roughly load-neutral behavior rather than as a claim about the base paper’s own (undocumented) construction.

We regard this list as a strength of the reconstruction methodology rather than a weakness: every entry reflects a place where the source material itself is ambiguous or incomplete, made explicit rather than resolved by assumption.

VII. CONCLUSION

We built a shared three-phase Newton–Raphson engine and compared direct, SVD/Moore–Penrose, rank-revealing QR, and Tikhonov-regularized linear solves across four independently reconstructed and cross-validated IEEE/PG&E test feeders. The central result is that no single method dominates: the same undamped, exact minimum-norm step that the base paper’s pseudo-inverse approach relies on oscillates rather than converges on a feeder whose floating transformer secondary spans the entire downstream network (IEEE 13-bus) at realistic load, while the identical Tikhonov regularization that rescues that case fails outright on a feeder where SVD and RRQR converge cleanly (IEEE 37-bus). A controlled symmetrical-component investigation shows that load imbalance on the 13-bus feeder’s topology concentrates almost entirely in the zero-sequence voltage component, offering a structural, partially explanatory account of that feeder’s characteristic elevated error. Finally, this project’s own development process demonstrates the practical value of independent cross-validation: two real, independent per-unit modeling bugs were caught only by validating small, well-conditioned sub-circuits against an external solver before trusting comparisons on the full, already-ill-conditioned systems — a discipline we recommend as a matter of course for any project reconstructing power-system test cases from published, and inevitably sometimes under-specified, sources.

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This work reproduces and extends the numerical methodology of Jang, Kim, and Kim [1]. Test feeder data are drawn from the IEEE PES Distribution System Analysis Subcommittee’s public test feeder repository [8] and the MATPOWER project [9], originally from Baran and Wu [10]. Independent validation used OpenDSS [11] and pandapower [12].

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